

Part 1A:

Testing 1

```
Hi there, please enter your first dna sequence: atgc
Thank you entering your first dna sequence!
Now, please enter your second dna sequence: atgc
Thank you entering your second dna sequence
This is your new DNA sequence: ATGCATGC
Length of your new DNA: 8
Reverse Complement of your DNA: GCATGCAT
Length of new RNA: 15
RNA: AUGCAUGCAAAAAAA
```

Testing 2:

```
Hi there, please enter your first dna sequence: actgtgtcagtcagtcg
Thank you entering your first dna sequence!
Now, please enter your second dna sequence: ttttgggtagctacgat
Thank you entering your second dna sequence
This is your new DNA sequence: ACTGTGTCAGTCAGTCGTTTTGGGTAGCTACGAT
Length of your new DNA: 34
Reverse Complement of your DNA: ATCGTAGCTACCCAAAACGACTGACTGACACAGT
Length of new RNA: 41
RNA: ACUGUGUCAGUCAGUCGUUUUGGGUAGCUACGAUAAAAAAA
```

Testing 3: To prove my while loop works

```
Hi there, please enter your first dna sequence: 1
oops..! try again
Hi there, please enter your first dna sequence: ATGC
Thank you entering your first dna sequence!
Now, please enter your second dna sequence: !
oops..! try again
Hi there, please enter your first dna sequence: A1
oops..! try again
Hi there, please enter your first dna sequence: ATGC
Thank you entering your first dna sequence!
Now, please enter your second dna sequence: a1
oops..! try again
Hi there, please enter your first dna sequence: ATGC
Thank you entering your first dna sequence!
Now, please enter your second dna sequence: ATGC
Thank you entering your second dna sequence
This is your new DNA sequence: ATGCATGC
Length of your new DNA: 8
Reverse Complement of your DNA: GCATGCAT
Length of new RNA: 15
RNA: AUGCAUGCAAAAAAA
```

Part 1B

Testing 1

```
Please enter value of a: 1
Please enter value of b: 2
Please enter value of c: 3
Please enter value of d: 4
please enter value of e: 5
You have entered: [1, 2, 3, 4, 5]
Mean: 3.0 (total = 15 , number of variables = 5)
Variance: 2.0
Standard deviation: 1.4142135623730951
```

Testing 2

```
Please enter value of a: 15
Please enter value of b: 30
Please enter value of c: 45
Please enter value of d: 20
please enter value of e: 35
You have entered: [15, 30, 45, 20, 35]
Mean: 29.0 (total = 145 , number of variables = 5)
Variance: 114.0
Standard deviation: 10.677078252031311
```

Testing 3

```
Please enter value of a: 11
Please enter value of b: 12
Please enter value of c: 13
Please enter value of d: 14
please enter value of e: 15
You have entered: [11, 12, 13, 14, 15]
Mean: 13.0 (total = 65 , number of variables = 5)
Variance: 2.0
Standard deviation: 1.4142135623730951
```

Part 1C

Testing 1

```
Enter side 1: 3
Enter side 2: 3
Enter side 3: 3
Entered values: [3, 3, 3]
Sorted values: [3, 3, 3]
Triangle is equilateral
```

Testing 2

```
Enter side 1: 2
Enter side 2: 3
Enter side 3: 10
Entered values: [2, 3, 10]
Sorted values: [2, 3, 10]
Triangle is impossible
```

Testing 3

```
Enter side 1: 2
Enter side 2: 2
Enter side 3: 1
Entered values: [2, 2, 1]
Sorted values: [1, 2, 2]
Triangle is isoceles
```

Testing 4

```
Enter side 1: 4
Enter side 2: 5
Enter side 3: 3
Entered values: [4, 5, 3]
Sorted values: [3, 4, 5]
Triangle is right angled
```

Testing 5

```
Enter side 1: 7
Enter side 2: 8
Enter side 3: 9
Entered values: [7, 8, 9]
Sorted values: [7, 8, 9]
Triangle is scalene
```

Testing 6

```
Enter side 1: -1
Enter side 2: 9
Enter side 3: 3
Invalid: All of the sides need to be within the range 1-1000
```

Part 1D

```
Distinction (mean=84.5): 100 99 99 96 80 79 78 78 78 76 76 75
Merit (mean=64.33): 68 67 66 65 60 60
Pass (mean=52.29): 59 56 56 55 50 45 45
Fail (mean=18.0): 0 0 10 12 21 24 24 25 31 33
Errors in position: 5 7 36
```

The end